



Cell Collective:

Enabling accessible and collaborative construction and analysis of comprehensive and annotated models

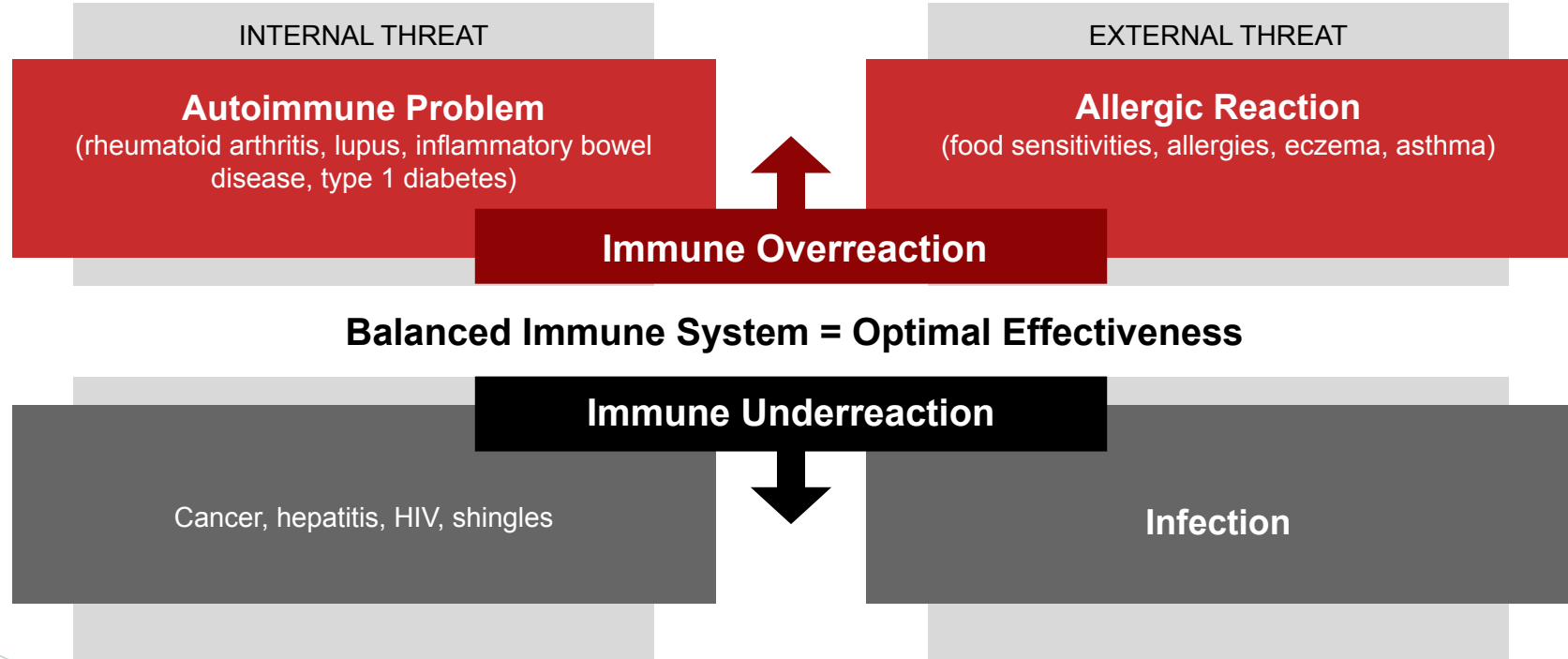
Tomas Helikar

www.cellcollective.org || @helikarlab || thelikar2@unl.edu

Eberhard Professor
Director, Digital Twin Innovation Hub
University of Nebraska at Lincoln



Our Interest: A Balanced Immune System



Everyone's Immune System Baseline "Status" is Different



Genetics



Pathogens



Vaccines



Health



Disease



Interventions



Behavior



Environment



Age

0

30

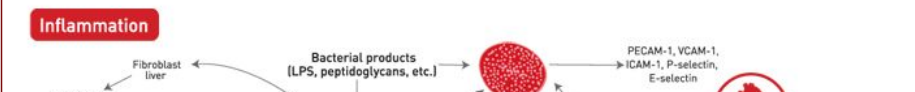
60

90

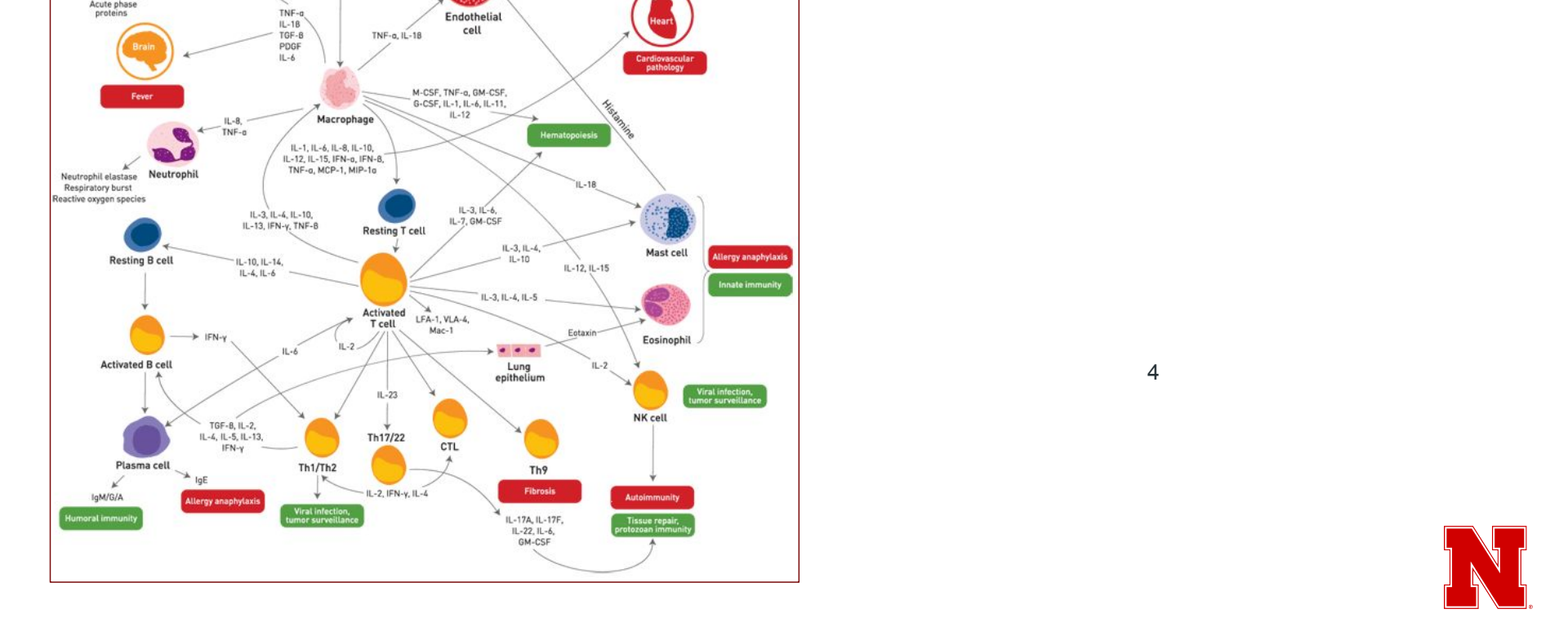
Immune Response,
Development, Diversity

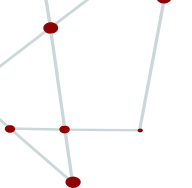
Inflammation
Immune Aging

Immune system is complex

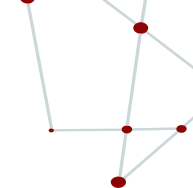


The diagram illustrates the inflammatory response. It shows a 'Fibroblast liver' cell on the left, which releases 'Bacterial products (LPS, peptidoglycans, etc.)'. These products interact with a red, textured cell (likely an endothelial cell) on the right. This interaction triggers the expression of adhesion molecules: 'PECAM-1, VCAM-1, ICAM-1, P-selectin, E-selectin'. A red blood cell is shown on the far right, with an arrow indicating its movement towards the site of inflammation.





Immune system is complex



How do we treat immune diseases today?

Whack a Mole!



Immune-related diseases are complex, costly, and chronically underserved

Immune system is the body's second most complex "organ"—dynamic, evolving, and patient-specific.

400+

Immune-related conditions—from cancer to sepsis—touch majority of therapeutic areas.

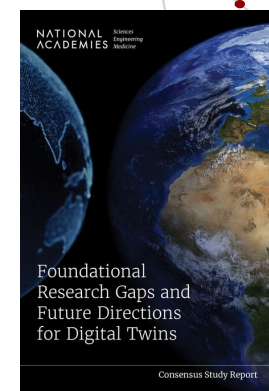
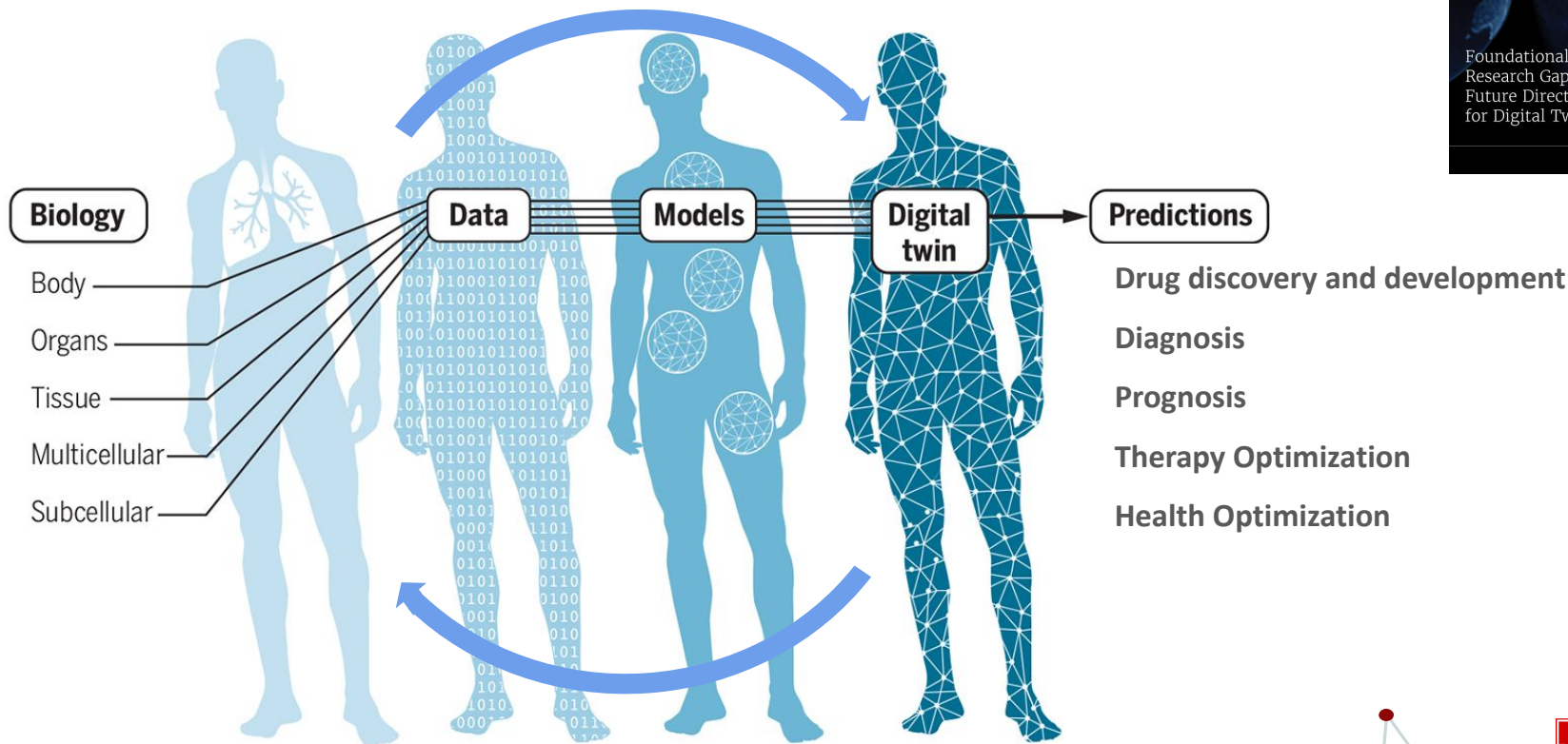
\$100B+

annual US burden for 50M patients - higher than diabetes.

40%

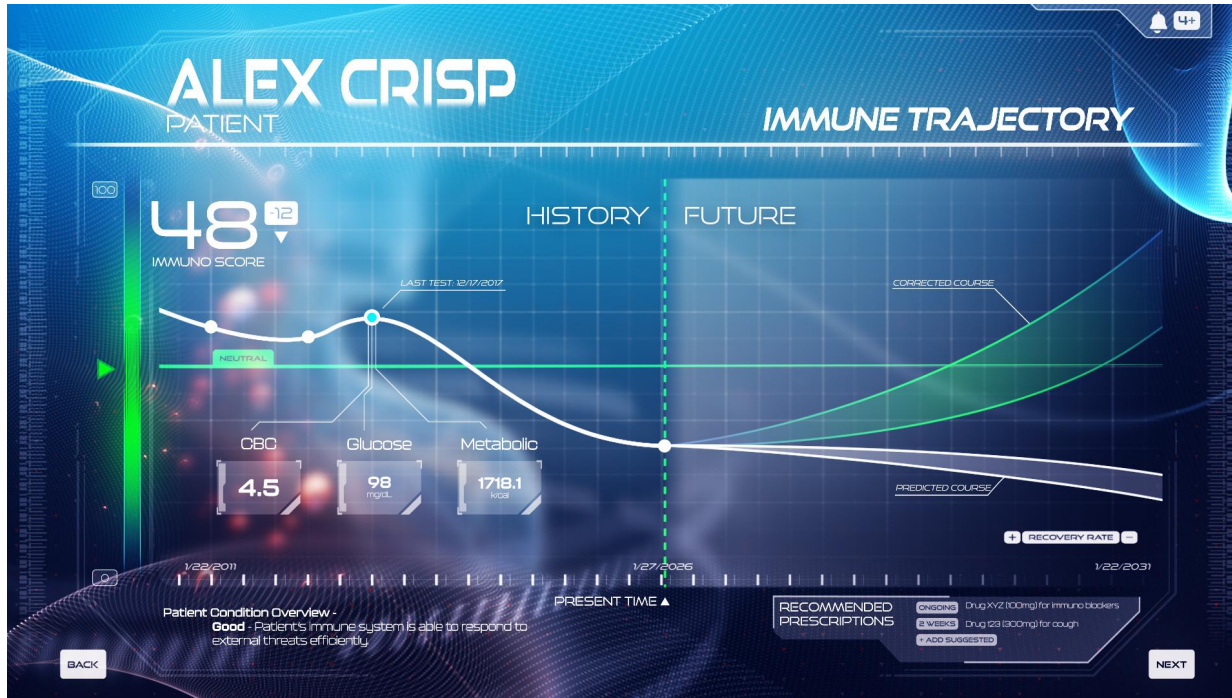
of patients fail first-line therapies, forcing costly trial-and-error care.

A digital twin for every immune system



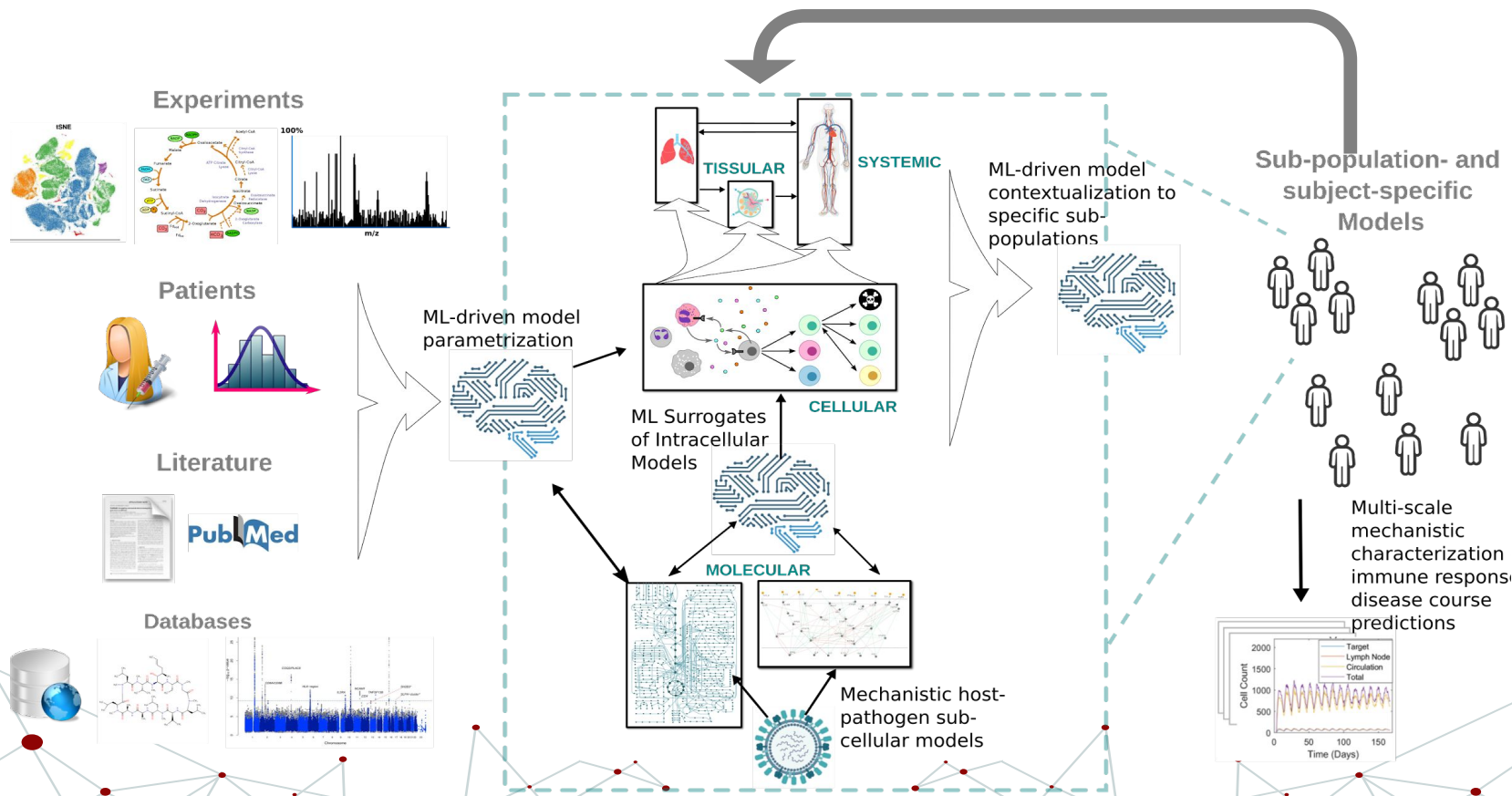
A digital twin for every immune system

- Develop a Scalable and Patient-Ready Immune Digital Twin Infrastructure
- Characterize and Quantify Immune System Health Across Scales
 - Identify multi-scale biomarkers / features that are predictive of immune system propensity to mount an effective response
- Enable personalized immune health prediction, forecasting, and optimization



Our Approach - Multiscale mechanistic - AI/ML Model

Patient-model synchronicity





Cell Collective: <https://cellcollective.org>



 Cell Collective

Student

Teacher

Researcher

Build large-scale models of biological systems, validate, simulate, perturb, and analyze data



Build Models

Collaboratively construct and annotate comprehensive mechanistic models



Simulate

Interactively simulate and analyze biological behaviors using a variety of built-in tools



Distribute

Share, distribute, and re-use models for and within the scientific community

Get Started as Researcher

Our goal:



1. ELIMINATE THE BAR

And Make Modeling **Accessible** to ANYONE



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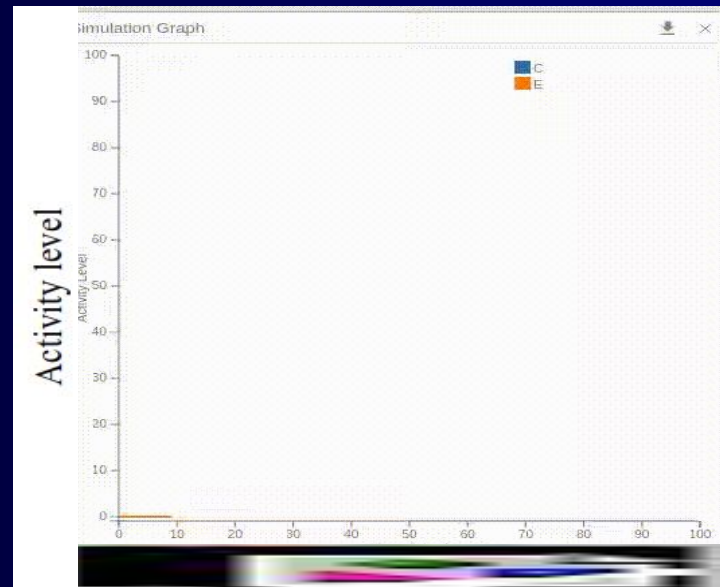
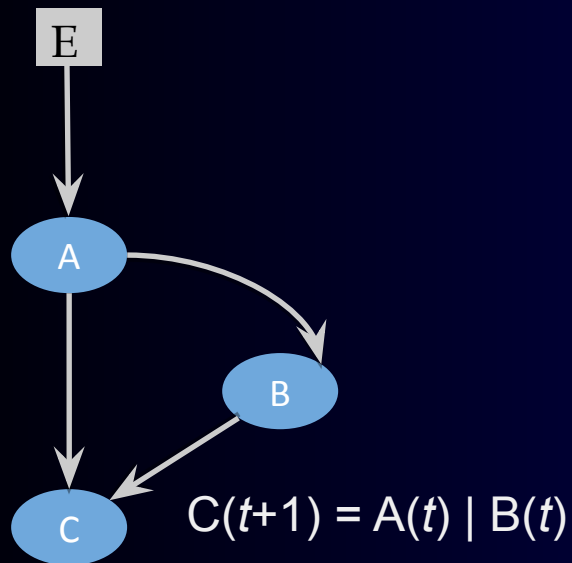
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Get Started as Researcher



Logical Models in Cell Collective



- Qualitative logic, i.e., “Proteins A or B activates protein C”
- (Kinetic) parameter independent
- Efficient simulations
- Delays (i.e., update after X steps of receiving an activation signal)
- Thresholds (i.e., update after X number of activating signals)
- Knowledge Base for detailed annotation

Constraint-based Models

- Flux Balance Analysis
- Flux Variability Analysis
- Custom Objective Function
- Context-specific Model Creation
- Define Gene Regulation
- Multi-layer Network Navigation
- Knowledge Base for detailed annotation

a Genome-scale metabolic reconstruction



b Mathematically represent metabolic reactions and constraints



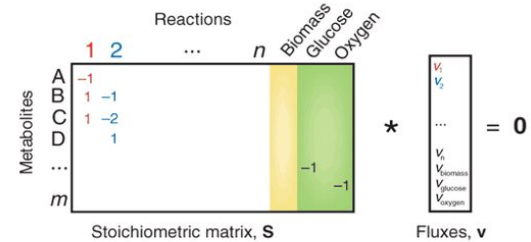
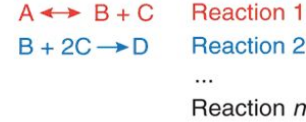
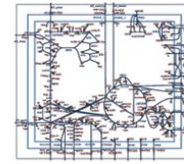
c Mass balance defines a system of linear equations



d Define objective function
($Z = c_1 \cdot v_1 + c_2 \cdot v_2 \dots$)

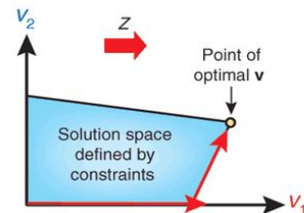


e Calculate fluxes that maximize Z



$$\begin{aligned}
 -v_1 + \dots &= 0 \\
 v_1 - v_2 + \dots &= 0 \\
 v_1 - 2v_2 + \dots &= 0 \\
 v_2 + \dots &= 0 \\
 \text{etc.}
 \end{aligned}$$

To predict growth, $Z = v_{\text{biomass}}$





Model Building

 **cellcollective**
research

Bacterial Chemotaxis
version 1.0

Tomas Helikar



[Model](#) | Simulation | Analysis | Network Analysis | Knowledge Base | Description

Graph
Layout: New Layout 1

Internal Components

External Components

Regulatory Mechanism

Positive Regulators	Negative Regulators

Knowledge Base



Multi-layer model visualization and navigation

Multi-scale model navigation

ccNetViz: a WebGL-based JavaScript library for visualization of large networks

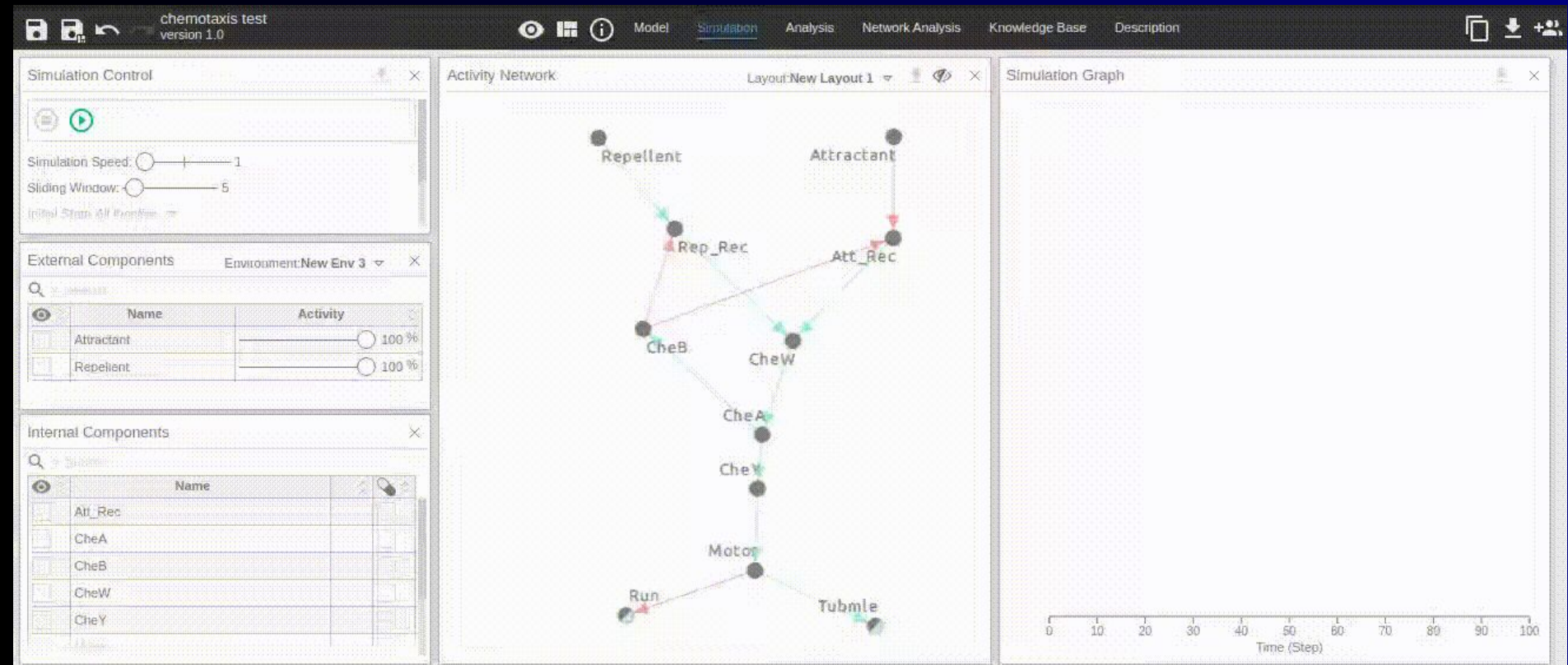
Ales Saska, David Tichy, Robert Moore, Achilles Rasquinha, Caner Akdas, Xiaodong Zhao, Renato Fabbri, Ana Jeličić, Gaurav Grover, Himanshu Jotwani, Mohamed Shadab, Resa M Helikar, Tomáš Helikar 

Bioinformatics, Volume 36, Issue 16, 15 August 2020, Pages 4527–4529,
<https://doi.org/10.1093/bioinformatics/btaa559>

Published: 09 June 2020 [Article history](#) ▼

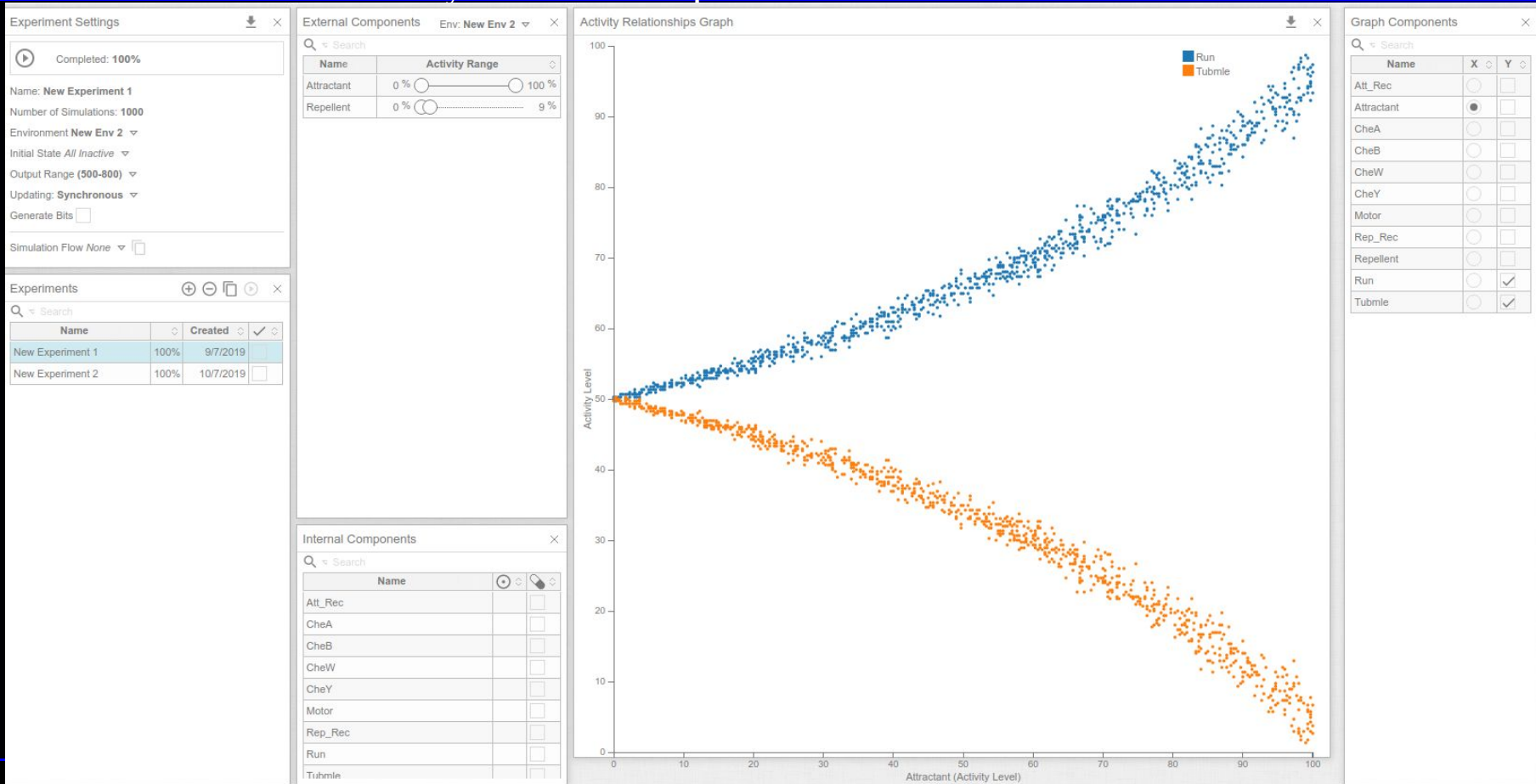


Model Simulations & Analysis: Real-Time Simulations





Model Simulations & Analysis: Dose-response / Titrations





Model Simulations & Analysis: Environment Sensitivity Analysis

bacterial chemotaxis version 1.0

Model Simulation Analysis Network Analysis Knowledge Base Description

Experiments

Q Search

Name	Created	✓
When do I run the most?	100% 9/6/2019	

External Components

Env: Default

Q Search

Name	Activity Range
Attractant	0 % 100 %
repellent	0 % 100 %

Component sensitivity for Run

Correlation

Component	Correlation
Attractant	0.9
repellent	-0.8

Experiment Settings

Completed: 100% Elapsed: 0s

Name: **When do I run the most?**

Number of Simulations: 1000

Environment Default

Initial State All Inactive

Output Range (500-800)

Updating: Synchronous

Generate Bits

Internal Components

Q Search

Name	☉	☒	↕
CheB			
CheW			
CheY			
Motor			
Rep_Rec			
Run			↑
Tubmle			

Environment Sensitivity

Environment	Sensitivity
Attractant	95
repellent	5



Model Simulations & Analysis: Feedback loop analysis

bacterial chemotaxis
version 1.0

ModelSimulationAnalysisNetwork AnalysisKnowledge BaseDescription

Tomas Helikar

Feedback Loops List

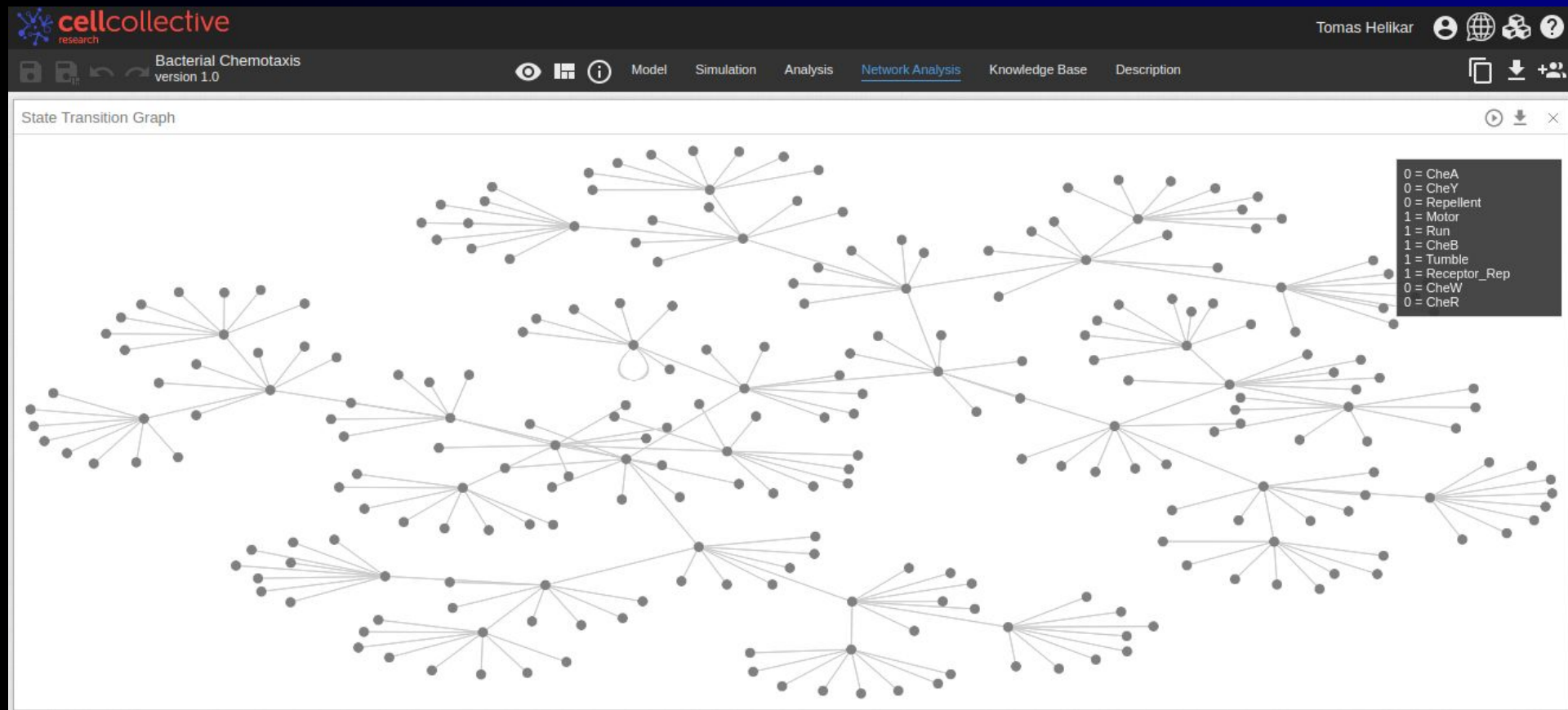
id	Feedback Loops
0	Rep_Rec -> CheW -> CheA -> CheB -> Rep_Rec
1	CheW -> CheA -> CheB -> Att_Rec -> CheW
2	CheW -> CheA -> CheB -> Rep_Rec -> CheW
3	Att_Rec -> CheW -> CheA -> CheB -> Att_Rec

Graph

Layout: Default



Model Simulations & Analysis: State Space Analysis



Our goal:



1. ELIMINATE THE BAR

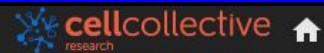
And Make Modeling **Accessible** to ANYONE

2. UNVEIL THE MECHANISM

Annotate and Explain



Cell Collective: Models annotation and transparency



Tomas Helikar



IGVH mutations in chronic lymphocytic leukemia.



[Knowledge Base](#)

[Simulation](#)

[Analysis](#)

[Model](#)

[Description](#)



Components

Search

Name
ADM
AEBP1
AFF1
AICDA
AKAP12
AKT3
ALOX5
ANXA2
APLP2
APOBEC3G
APP
BLNK
BMI1
CASP3
CAV1
CCL5
CCND2
CD27
CD63
CD69
CD70

Knowledge Base ADM

Description

- Adrenomedullin
- Gene ID: 133
- ADM and PAMP act as hypotensive and vasodilator agents. Both mediate the loss of plasma volume in the brain and pituitary gland and inhibit aldosterone secretion. (Alvarez-Silva MC, 2015)
- Gene: ADM
- UniProt ID: P35318

Regulatory Mechanism Summary

HSP90AA1 or TNF in conjunction with HSP0AA1 activates ADM. (Hofbauer KH, 2002)
(Masoud GN, 2015)

Upstream Regulators

HSP90AA1

HSP90AA1 indirectly upregulates ADM expression. (Masoud GN, 2015)

TNF

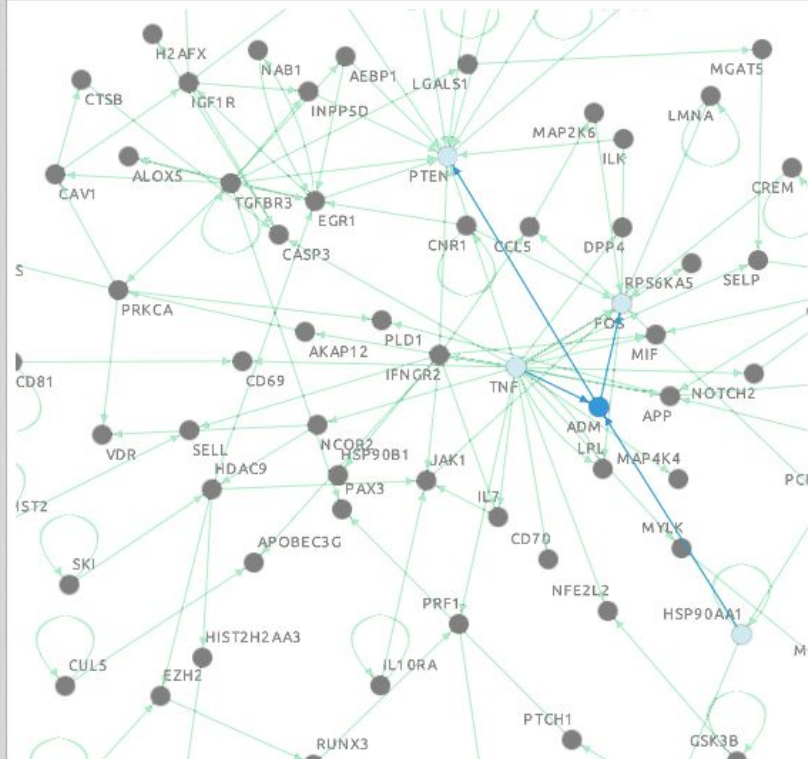
Cells treated with TNF showed upregulation of ADM mRNA. (Hofbauer KH, 2002)

References

- Hofbauer KH, Schoof E, Kurtz A, and Sandner P. Inflammatory cytokines stimulate adrenomedullin expression through nitric oxide-dependent and -independent pathways. Hypertension 2002 Jan 39; (1) 161-7. pmid:11799096
- Masoud GN and Li W. HIF-1 α pathway: role, regulation and intervention for cancer therapy. Acta Pharm Sin B 2015 Sep 5; (5) 378-89. pmid:26579469
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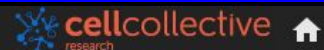
Reference Graph

Layout: Default





Cell Collective: Models annotation and transparency



Tomas Helikar



IGVH mutations in chronic lymphocytic leukemia.



[Knowledge Base](#)

[Simulation](#)

[Analysis](#)

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[Description](#)



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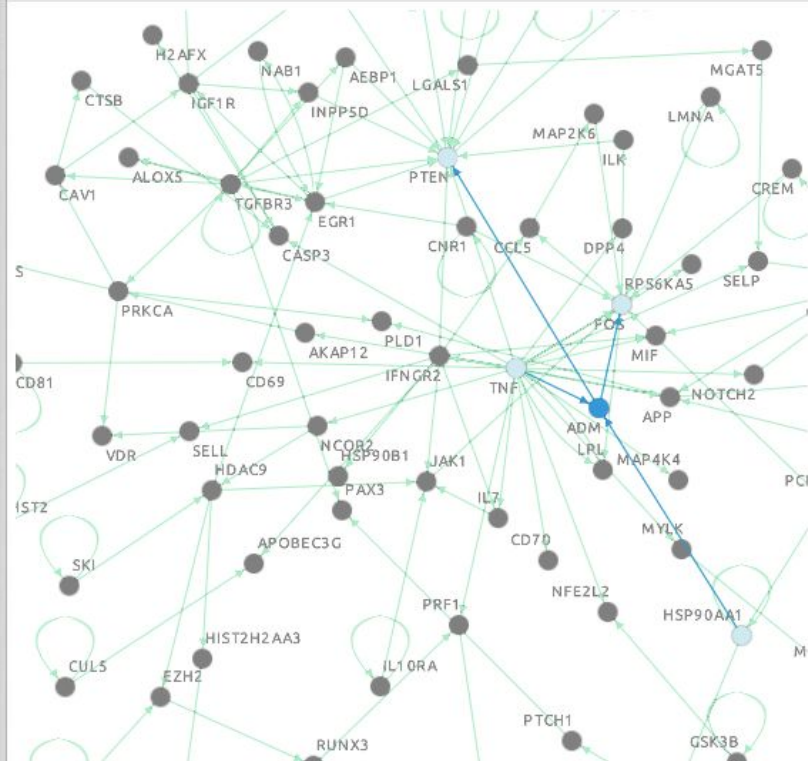
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Reference Graph

Layout: Default



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2. UNVEIL THE MECHANISM

Annotate and Explain

3. COLLABORATE

To **Construct** More Detailed, Comprehensive
and Accurate Models

To Use Models to **Discover**



Collaboration, Repository, and Model Access



SBML Qual Paper Final Model demo



Knowledge Base

Simulation

Analysis

Model

Description

Share with Collaborators



Search

Email

audrey.crowther@huskers.unl.edu



Publish your Model



By publishing this model, it will become visible to all Cell Collective users who will be able to simulate the model and create their own copy to help expand on it in a collaborative fashion (only the original model creator and the users with whom the creator shared the model can continue to directly edit the published model).

PUBLISH

Model cannot be published.

Publishing requirements

- ✗ *Model has to have been included in at least one peer-reviewed publication.*
There are no publications associated with this model (You can add publications under the Description page).
- ✗ *Each model component should be annotated in the knowledge base in an effort to make models more reproducible, transparent and accessible to the community.*

Components without knowledge base information:

akt ap1 ask1 cjun creb egf egfr erk ex gsk3 ikb ikk jnk map3k1 map3k7 mek
mkk4 mkk7 nfkb nik p38 p90rsk ph pi3k raf1 ras sos tnfa tnfr traf2

- ✓ *User profile of the model owner has to be filled.*

Experiments Publishing



Search

Name

TNFalpha titration

Shareable Links



Search

Link

https://cellcollective.org/#a2b73520-ffc4





Collaboration, Repository, and Model Access - SBML import



Tomas Helikar



rheumatoid arthritis map
version 1

Model Simulation Analysis Network Analysis Knowledge Base Description



Internal Components

Q Search

Name		
ABL1	2	0
ACT1	1	0
Actin Based Motility, Cell Migration	1	0
active_cAMP	1	1
activated_H_sub_2_endsub_O_sub_2	1	0
ADAMTS-9	1	1
ADCY7	1	1
ADM	3	0
AKT2	2	1
Apoptosis	9	0

External Components

Q Search

Name
ABL1
ACT1
ADAMTS-9

Regulatory Mechanism ABL1

Positive Regulators Drop Component

Negative Regulators Drop Component

ABL1

Conditions

Condition
If/When EPHA4 is Active

SubConditions

SubCondition
If/When CRKL is Inactive

Knowledge Base ABL1

CRKL

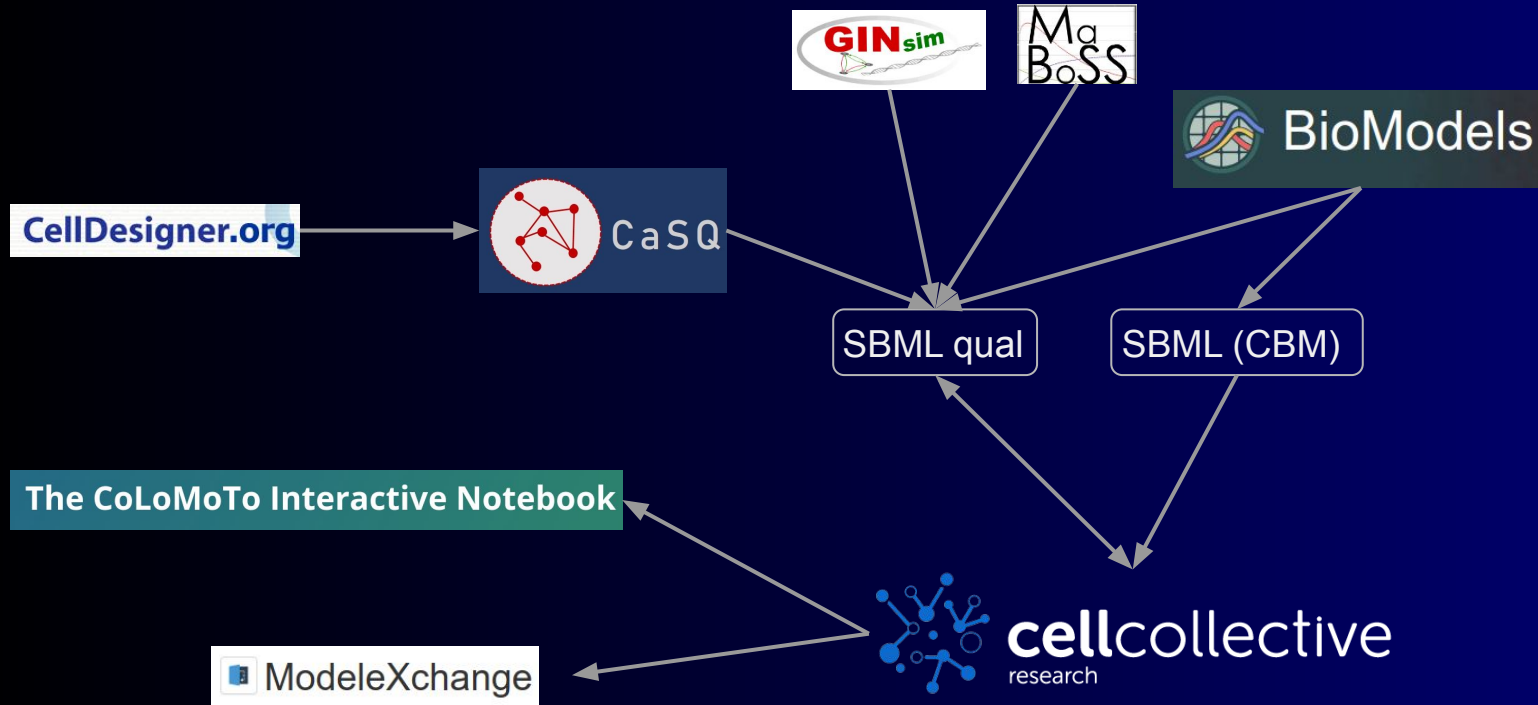
EPHA4

References

- Lindberg J, af Klint E, Catrina AI, Nilsson P, Klareskog L, Ulfgren AK, and Lundberg J. Effect of infliximab on mRNA expression profiles in synovial tissue of rheumatoid arthritis patients. Arthritis Res Ther 2006; (6) R179. PMID: 17134501



Cell Collective - Ecosystem & Collaboration



Cell Collective Model Repository

Support for multi-method modeling techniques and sub-models:

- Signal Transduction & Gene regulation (logical models)
- Kinetic models (ODEs)
- Metabolism (constraint-based models)



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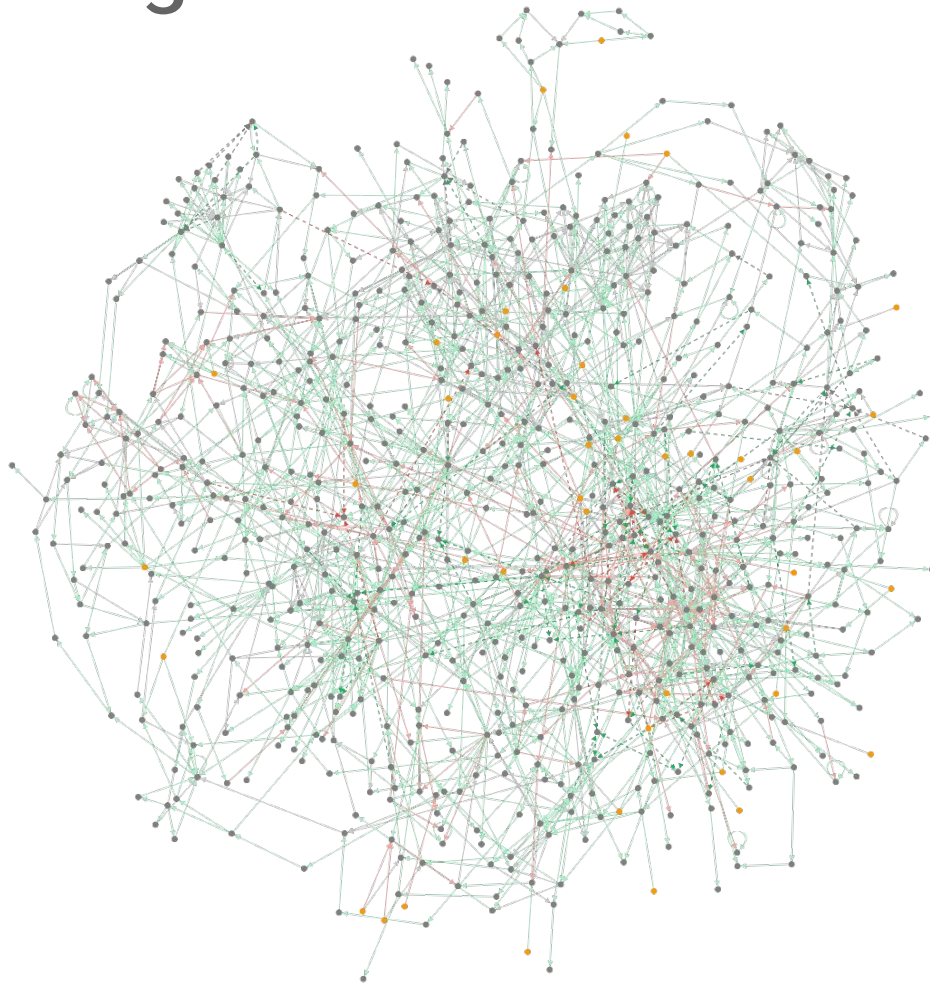
To **Construct** More Detailed, Comprehensive
and Accurate Models

To Use Models to **Discover**

4. DON'T REINVENT THE WHEEL

And Allow Others **Re-Use** and **Reproduce**
your Work

But, how do I get started with this mess ...?



Custom Interactive Model Guides

Simulating the Behavior of Cellular Respiration (163274)

File Insert Edit Workspace Reports Help

Overview **Start Here**

Page 1 Cellular Respiration Page 2 Cellular Respiration Page 3 No Glucose Page 4 Pyruvate Processing Inhibited Page 5 Pyruvate Processing Inhibited

1. Simulation of the cellular respiration system behavior

The purpose of this simulation is to explore how the inputs and outputs of each process of cellular respiration impact the activity of the entire system. We'll walk you through how to set the simulation controls, and then ask you to describe the model behavior once you play the simulation.

- Within the Activity Network panel, click the following components to change them grey-blue; this two-toned component color means they will be visible within the Simulation Graph, once the simulation is played.
 - glycolysis
 - pyruvate_processing
 - citric_acid_cycle
 - electron_transport_chain
- Within the External Components panel, set all External components to 100%. This means that all external stimuli are available and active.
- Set the Sliding Window to "100".
- Start the simulation by clicking on the *play* button in the Simulation Control panel; pause after ~80 time steps.

Activity Network

Layout: Cellular Respiration

Simulation Control

Simulation Speed: 1
Sliding Window: 1
Initial State: All Inactive
Updating: Synchronous
Simulation Flow: None

Simulation Graph

External Components

Environment: Default

Name	Activity
1.glucose	0 %
2.NAD+	0 %
3.ADP	0 %
4.ATP	0 %
5.FAD+	0 %
6.oxygen	0 %
fatty_acid_metabolism	0 %

Custom Interactive Model Guides

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Custom Interactive Model Guides

Exploring Prokaryotic Gene Regulation in the Lac Operon (118235) - changes not saved

File Insert Edit Workspace Help

Overview 1. Build the Glucose Path 2. Build the Lactose Path 3. Predict Model Behavior 4. Simulate Behavior 5. Mutation Simulations 6. Strengths/Limitations

Step 1: Build Glucose Path 1.0 Step 2: Build Glucose Path 1.0 Step 3: Build Glucose Path 1.0 Step 4: Build Glucose Path 1.0 Step 5: Build Glucose Path 1.0 Step 6: Build Glucose Path 1.0

Model Building Questions 1

Feel free to navigate back to previous sections of the lesson to find the best answers. Save before switching steps so you will not lose any answers. You are also encouraged to use the connections in your model as a resource to help you answer the following questions when possible.

Q1. The only signal detected by the lac operon is environmental lactose. Participation Model Building No Objective T [] [] Points: 1 X

Options: +

☐ True

☒ False

Q2. The lac operon detects both glucose and lactose signals from the environment. Participation Model Building No Objective T [] [] Points: 1 X

Options: +

☒ True

☐ False

Participation Model Building

Discuss the lac operon as a prokaryotic gene regulation system and its overall importance for controlling lactose digestion T [] [] Points: 1 X

Q3. The lac operon is an inducible gene regulation system that controls the production of enzymes that digest lactose.

Options: +

☒ True

☐ False

Graph

Layout: New Layout 4

Glucose Lactose

Custom Interactive Model Guides

Exploring Prokaryotic Gene Regulation in the Lac Operon (118235) - changes not saved

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Options: +

☐ True

☒ False

Q2. The lac operon detects both glucose and lactose signals from the environment. Participation Model Building No Objective T [] [] [] Points: 1

Options: +

☒ True

☐ False

Participation Model Building

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Q3. The lac operon is an inducible gene regulation system that controls the production of enzymes that digest lactose. Options: +

☒ True

☐ False

Graph Layout: New Layout 4 + - [] [] [] [] [] [] [] []

Glucose Lactose

Custom Interactive Model Guides

Exploring Prokaryotic Gene Regulation in the Lac Operon (118235) - changes not saved

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Model Building Questions 1

Feel free to navigate back to previous sections of the lesson to find the best answers. Save before switching steps so you will not lose any answers. You are also encouraged to use the connections in your model as a resource to help you answer the following questions when possible.

Q1. The only signal detected by the lac operon is environmental lactose. Participation Model Building No Objective T [] [] [] Points: 1

Options: +

☐ True

☒ False

Q2. The lac operon detects both glucose and lactose signals from the environment. Participation Model Building No Objective T [] [] [] Points: 1

Options: +

☒ True

☐ False

Participation Model Building

Discuss the lac operon as a prokaryotic gene regulation system and its overall importance for controlling lactose digestion T [] [] [] Points: 1

Q3. The lac operon is an inducible gene regulation system that controls the production of enzymes that digest lactose.

Options: +

☒ True

☐ False

Graph

Layout: New Layout 4

Glucose

Lactose

Custom Interactive Model Guides

Overview

Start Here

Page 1
1.0

Page 2
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Page 3
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Page 4
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Page 5
1.0

Page 6
1.0

Submit Lesson
1.0

1. Model of light and dark reactions

On the right, a model of light and dark reactions in the process of photosynthesis has been built for you. Click on white space within the graph panel to de-activate the selection of any components/arrows (in case you already started exploring!).

What you should notice in this model is the following:

- There are two processes occurring: light-dependent reactions and the Calvin Cycle.
- Matter is cycling between these two processes: ADP and NADPH are used during the Calvin Cycle where ATP is reduced to ADP+Pi and NADPH is oxidized to NADP+. NADP+ and ADP+Pi are then recharged during light-dependent reactions, back to ATP and NADPH.
- In order for light-dependent reactions to function, not only are NADP+ and ADP+Pi necessary, but H₂O and sunlight are also required.
- In order for the Calvin cycle to function, not only are ATP and NADPH required, but CO₂ is also required.

Q1. Describe the energy

Assessment

General Skills

No Objective

T

☑

☰

Points: 1

×

molecules that are generated by the light-dependent reactions and the energy molecules that are generated by the Calvin Cycle.

Example answer: The light-dependent reactions produce oxygen, ATP (for energy storage), and NADPH (the reduced electron carrier). The Calvin cycle uses both ATP and NADPH to produce NADP+, ADP, and glucose. Both NADP+ and ADP are recycled and used in the light-dependent reactions.

graph

Layout: New Layout 1



Our goal:



1. ELIMINATE THE BAR

And Make Modeling **Accessible** to ANYONE

2. UNVEIL THE MECHANISM

Annotate and Explain

3. COLLABORATE

To **Construct** More Detailed, Comprehensive
and Accurate Models

To Use Models to **Discover**

4. DON'T REINVENT THE WHEEL

And Allow Others **Re-Use** and **Reproduce**
your Work



Cell Collective: <https://cellcollective.org>



 Cell Collective

Student

Teacher

Researcher

Build large-scale models of biological systems, validate, simulate, perturb, and analyze data



Build Models

Collaboratively construct and annotate comprehensive mechanistic models



Simulate

Interactively simulate and analyze biological behaviors using a variety of built-in tools



Distribute

Share, distribute, and re-use models for and within the scientific community

Get Started as Researcher

Our goal:

1. ELIMINATE THE BAR

And Make Modeling Attainable to
ANYONE

2. Give students the

opportunity

to practice, practice, and practice in
a **constructivist, inquiry-based,**
and scientifically authentic
fashion



Broad application of Cell Collective Modeling Lessons



Key Lesson Design Features

- Model-building and simulations
- Self-contained
- Simple, turn-key implementation
- Predict/observe/explain scenarios
- Modular implementation
 - Laboratory activity
 - Homework
 - In-class activity

Published Models (11) My Learning (1) My Models (293) Shared with Me (47)

Lac Operon Construction
Components: 0 Interactions: 0
Author: Nicholas Galt
Audience: Freshmen
Created: 6/20/2016 Updated: 6/20/2016

Glucose Homeostasis
Components: 0 Interactions: 0
Author: Nicholas Galt
Audience: Freshmen
Created: 6/19/2016 Updated: 6/19/2016

Regulation of Cellular Respiration
Components: 28 Interactions: 59
Author: Nicholas Galt
Audience: Freshmen
Created: 5/29/2016 Updated: 6/19/2016

G2 Checkpoint Regulation and the FA/ER
Components: 31 Interactions: 127
Author: Audrey Crowther
Audience: Sophomore
Created: 6/15/2016 Updated: 6/16/2016

Cell Cycle Regulation: The G1/S Transition
Components: 21 Interactions: 54
Author: Audrey Crowther
Audience: Sophomore
Created: 6/13/2016 Updated: 6/13/2016

Cell-to-Cell Communication Networks
Components: 29 Interactions: 60
Author: Audrey Crowther
Audience: Senior
Created: 6/3/2016 Updated: 6/10/2016

Regulation of CD4+ T Cell Differentiation
Components: 16 Interactions: 30
Author: Audrey Crowther
Audience: Senior
Created: 6/3/2016 Updated: 6/9/2016

Cell Cycle Regulation and Tumorigenesis
Components: 34 Interactions: 165
Author: Audrey Crowther
Audience: Freshmen
Created: 6/3/2016 Updated: 6/7/2016

Positive and Negative Feedback Loops
Components: 13 Interactions: 14
Author: Audrey Crowther
Audience: Freshmen
Created: 6/3/2016 Updated: 6/7/2016

~50 Modeling Lessons



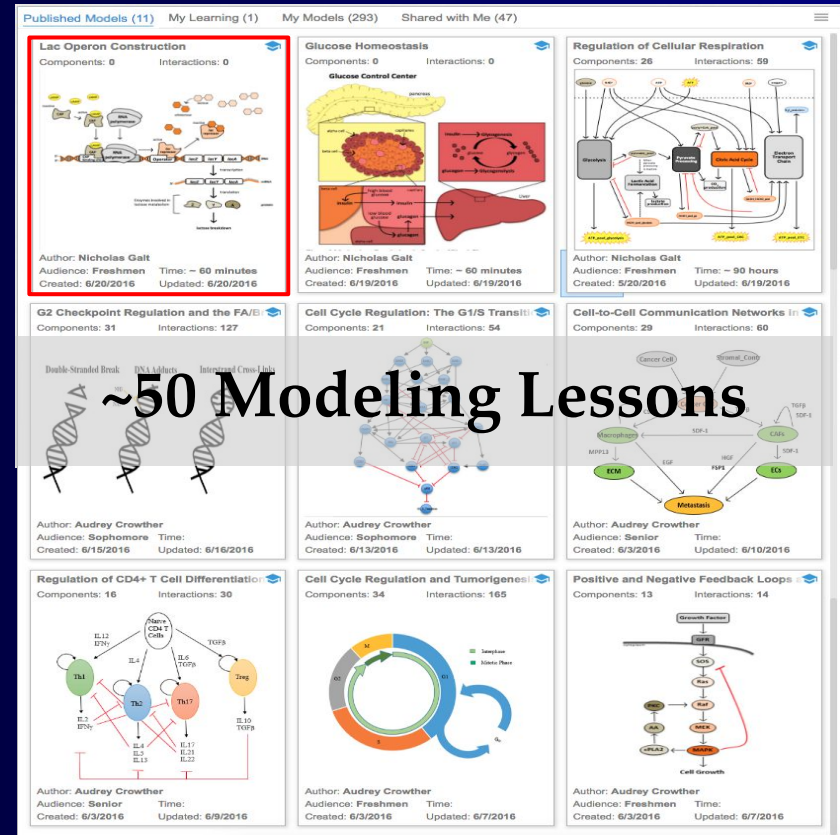
Broad application of Modeling Lessons

Topics

- Metabolism: Glycolysis, TCA cycle, ETC
- Cell respiration
- Glucose homeostasis
- Cell cycle
- Photosynthesis
- Gene regulation
- Food web dynamics

Courses (high school - graduate)

- Biochemistry
- Intro biology courses
- Immunology
- Cancer Biology
- *Many others*



Modeling lesson design

Build a model

Make prediction (*hypothesis generation*)

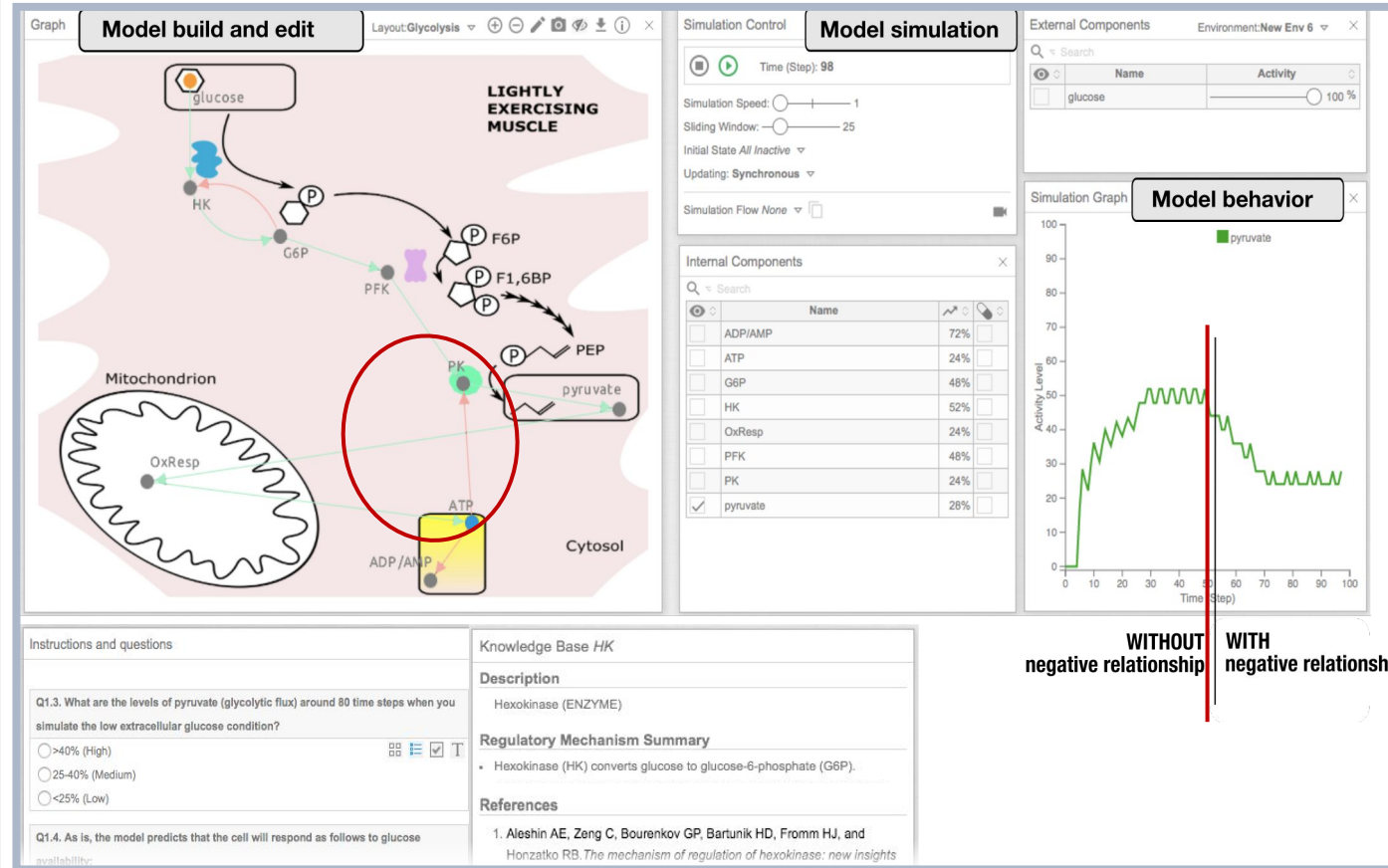
Justify prediction (*mechanistic thinking*)

Test prediction (*experimental design*)

Report simulation results (*data interpretation*)

Evaluate your prediction (*data interpretation*)

Refine prediction using simulation results (*mechanistic thinking, hypothesis generation*)





Cell Collective World-wide Adoption

Published Models (77) My Models (42) Shared With Me (281)

RECENTLY PUBLISHED

HIV-1 Interactions with T Cell Signalling Pathway
Version 1.0
Components: 138
Interactions: 138

T Cell Receptor Signaling
Version 1.0
Components: 101
Interactions: 158

Signaling Pathway for Butanol Production in Clostridium beijerinckii
NRRL B-598
Components: 100
Interactions: 100

Neurotransmitter Signaling Pathway
Version 1.0
Components: 16
Interactions: 22

Interactive Map

Author: C
Score: 45
Created: 5/13/2014

Author: A Fauré et. al.
Score: 132.8
Created: 5/13/2014

Author: S Li et. al.
Score: 106.2
Created: 6/19/2013

Author: S Klamt et. al.
Score: 98.9
Created: 2/9/2015

Author: J Saez-Rodriguez et. al.
Score: 92.9
Created: 6/23/2013

Cited: 0
Updated: 8/5/2018

Cited: 346
Updated: 7/19/2017

Cited: 307
Updated: 7/19/2017

Cited: 306
Updated: 1/22/2016

Cited: 220
Updated: 8/9/2019

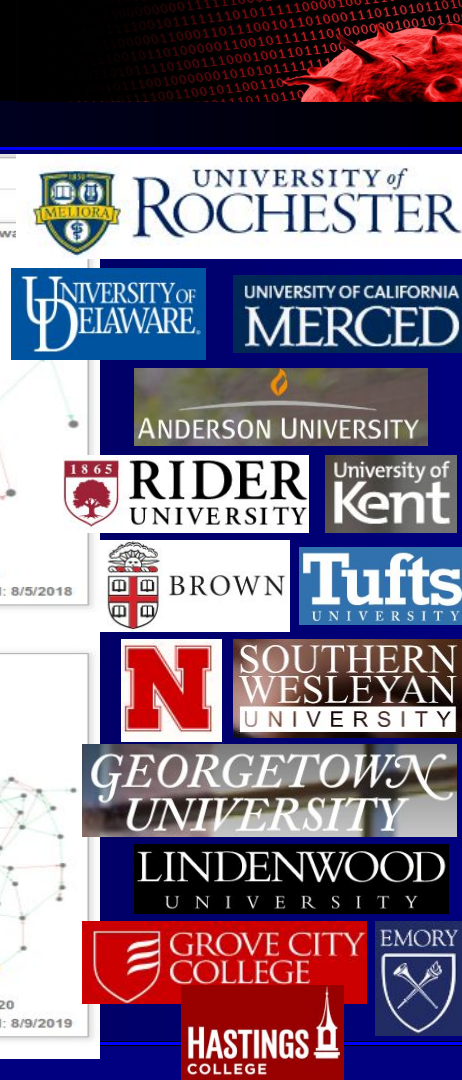
MOST PUBLISHED

Mammalian Signalling Pathway
Version 1.0
Components: 138
Interactions: 138

Mammalian Signalling Pathway
Version 1.0
Components: 138
Interactions: 138

Mammalian Signalling Pathway
Version 1.0
Components: 138
Interactions: 138

Mammalian Signalling Pathway
Version 1.0
Components: 138
Interactions: 138



Long-term vision: Digital Twins as Infrastructure

